CCAT Reconciliation in Snowflake

Detect, fix, and prove it: keeping our equipment records and Caterpillar's registry in sync

Cleveland Brothers · Data & Analytics

WHY IT MATTERS

The scoreboard this feeds: Trifecta and DNA

TRIFECTA

Caterpillar's data-quality score. A customer only counts as valid when ALL legs pass: CCID, company data, a valid contact, and 100% asset match on priority assets.

92.8%

asset match conformance (2025 baseline) — the leg our data governance team is focused on, and the one still maintained by hand today

Worth 15% of the dealer excellence scorecard

DNA — DATA NOTIFICATION ALERTS

Caterpillar flags our prioritized data issues — asset alerts and customer alerts — sends us the top ones by value, and grades how well we resolve them.

10%

of the scorecard rides on DNA resolution — and many of the asset alerts trace back to the same root cause: ownership records that do not match reality

Both metrics stand on the same ground: asset ownership records in CCAT. That is exactly what this project manages.

WHY IT MATTERS

This project is the asset-match engine

Machines missing from CCAT	get added, audited — they enter Caterpillar's population on the right customer
Machines on the wrong customer	get reassigned — the exact defect that breaks a customer's asset match
Machines other dealers claim	get reviewed transfers — closing the gaps no amount of data entry can fix
Manual upkeep becomes a pipeline	nightly detection, audited fixes, and alerts prevented at the source instead of resolved after they fire

One wrong machine fails its whole customer for Trifecta. Every machine this system fixes can flip an entire customer back to valid — the same work, scored twice: Trifecta up, alert queue down.

THE BIG PICTURE

The whole system on one slide

DETECT

A nightly, read-only sweep checks each machine against Caterpillar's live API and classifies it. It never writes anything to Caterpillar.

ACT

Audited Snowflake procedures do the fixes: adds, customer reassignments, and (reviewed) transfers. Every write previews first and refuses unsafe cases on its own.

PROVE

Two permanent tables: AUDIT records everything we did; HISTORY records every version of the data anyone ever produced. Nothing is ever edited or deleted.

The loop closes itself: every action's result is verified by the same sweep that found the problem — the system never marks its own homework.

DETECT

The nightly sweep: CCAT_DETECT_DISCREPANCIES

One live API search per machine
Compares Caterpillar's answer against our golden equipment view, field by field.

Smart about what it re-checks
A machine is only re-checked if it was never checked, our side changed, the last check errored, or the check has gone stale. Everything else is skipped for free.

Writes everything downstream
Fills the work queues, updates each machine's state, and snapshots any changed CCAT data into history. Returns a summary of every run.

~0.9s

per machine checked — one polite, paced API call

5,000

machines a night in about 75 minutes — full fleet coverage builds in weeks, then maintains itself

DETECT

What the sweep writes: queues, state, and memory

CCAT_MISSING

Machines with no record under our dealer code — the add and transfer pipeline. Keeps the other dealers' records for triage.

CCAT_NON_MATCHING

Field-level differences on machines we hold — one row per differing field, judged against our best-matching CCAT record.

CCAT_ERRORS

Failed checks. A failure is never recorded as a discrepancy — it lands here with Caterpillar's tracking ID for support.

CCAT_CHECK_STATE

One row per machine ever checked: verdict, ownership type, customer, pending flags — the sweep's memory and the dashboard's engine.

CLASSIFY

Every machine gets a phase, an action, and a why

Done	Record is right. Nothing to do.
Phase 1 — Add	Not in CCAT anywhere. Adding affects nobody. Batchable.
Phase 2 — Quiet fix	Other records exist but none are exclusive. Completes without notifying anyone.
Phase 3 — Fix ours	We own the record: backfill missing values, or move it to the correct customer.
Phase 4 — Review	Transfers and shared ownership. A person decides, about ten per shift.
Info only	Model differs but CCAT is the authority — reported, no action queued.

One query answers it all

The phase dashboard shows, per machine:

PHASE · ACTION · WHY

plus both customer numbers, ownership type, and which other dealers are involved. The rollout plan, applied to live data.

The decision key, animated

One machine. A few questions.

How the system decides what is safe to do

83 seconds, narrated

Three real machines walk the decision tree:

a safe add,
a quiet add,
and a reviewed transfer.

Click to play.

Acting: every write goes through an audited procedure

CCAT_ADD_ASSET

CCAT_REASSIGN_CUSTOMER

- Look up the machine by serial number, pull everything else from our records
- Search CCAT live before acting — never write blind
- Refuse anything outside their lane: existing records, other owners, attachments, known-bad serials
- Dry-run by default — show exactly what would be sent, send nothing
- On execute: one API call, one audit row, one history snapshot

Payload rule

Where CCAT already has a record, its model and year are the versions Caterpillar recognizes — our requests reuse them. Our data only goes to Caterpillar for machines it has never seen.

Raw API tools still exist

The low-level add/expire/transfer procedures are diagnostics only. They are not audited, and when access rolls out, other users will only ever be granted the wrappers.

PROVE

CCAT_AUDIT: what we did, forever

Every executed or failed action writes one row:

- What: action type, phase, machine, model, ownership type
- Customer before → customer after (the reassignment story)
- The exact payload sent, and Caterpillar's full response
- Caterpillar's tracking ID — their support can trace any call
- A snapshot of CCAT taken just before we acted (the undo reference)
- Who, when, and which shift — feeds the per-shift reporting

Append-only by permission

The tooling role can INSERT and SELECT — nothing else. Audit rows cannot be edited or deleted, even by accident.

Kept forever. This table is the compliance trail.

PROVE

CCAT_HISTORY: what the data was, at every point

Every distinct version, any author

Each check fingerprints Caterpillar's full answer for the machine. New fingerprint = the full snapshot is stored. Same fingerprint = nothing.

Cheap by design

A machine checked fifty times with no change stores one row, not fifty. Built entirely from data the sweep already fetches — zero extra API calls.

Absence counts too

“No records at all” is itself a version — a machine appearing in, or vanishing from, CCAT leaves a trail.

1 row

per real change in Caterpillar's data, no matter how many times we look.

Append-only and kept forever, same as audit.

PROVE

Two tables, one powerful difference

AUDIT

What WE did.
Actions, payloads, outcomes, tracking IDs — signed, dated, attributable.

HISTORY

What the data WAS.
Every observed version of every machine's CCAT picture, whoever changed it.

History minus audit = external changes. A version change with no audit row nearby means Caterpillar, another dealer, or a portal edit changed the data — the class of change that used to be invisible.

RUN IT

The operating rhythm

Nightly

The sweep checks its batch, refreshes verdicts, and grows coverage. Read-only, automatic.

Weekly

Batched safe adds and quiet fixes run under volume caps. A scoreboard snapshot tracks the backlog shrinking.

Every shift

About ten reviewed decisions: transfers, inbound claims, judgment calls — audited as they happen.

What leadership sees, weekly:

- Customer assignment rate — the headline: % of machines on the correct customer in CCAT
- Coverage and match rate — how much of the fleet is verified, and how much agrees
- Backlog by phase, resolutions per shift, and transfers in flight

GUARDRAILS

The rules that keep it safe

Dry-run first, always

Every write shows its exact payload and sends nothing until deliberately executed.

No bulk claims, ever

Blanket adds would fire transfer requests at real dealers and can silently overwrite our own records.

Shared customers excluded

When another dealer serves the same customer, the machine routes to coordination — never a claim.

CCAT owns the model

We never push our model names over Caterpillar's — their versions are what Cat systems recognize.

Production awareness

Caterpillar has no test environment. Every write is treated as real, because it is.

Compliance tables are sacred

Audit and history are append-only by permission and kept forever — off the reset list permanently.

STATUS

Where it stands

Working today

- Detection sweep, proven at thousand-machine scale
- Phase dashboard: phase, action, and why per machine
- Audited add + customer reassignment procedures — real adds executed in production
- Audit + history compliance layer
- Full documentation and a versioned SQL reference

Next

- Schedule the sweep as a nightly Snowflake task
- Batch runner for the safe-add pile, with volume caps
- Backfill and transfer wrappers (same audited pattern)
- Reader/writer access roles once rollout is approved
- Review workflow at ~10 resolutions per shift

The system sorts every machine.

People decide only the ones that need judgment.

Code: SQL Reference folder (files 40–49) · Docs: 10–14 · Dashboard: V_CCAT_PHASE_DASHBOARD